Activity -7

The Water Jug Problem

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1. Introduction

Artificial Intelligence (AI) is the branch of computer science that aims to make computers solve problems in a way that appears intelligent. A very common AI technique is search: the computer tries one action after another, keeping track of what it has already tried, until it finds a sequence of actions that gives the required result.

To study and test search techniques, AI courses use small, clearly defined puzzles known as "toy problems". The Water Jug Problem is one of the oldest and best known. Even though it uses only two jugs and a tap, it teaches all the important ideas used in much larger real problems: states, actions, goals and search.

This report explains the problem from the beginning, converts it into a form a computer can solve, runs a systematic search, and gives the complete step-by-step solution in simple language.

2. Problem Statement

You are given two empty jugs:

- One jug that can hold exactly 4 gallons (the 4-gallon jug).
- One jug that can hold exactly 3 gallons (the 3-gallon jug).
- A pump (tap) with an unlimited supply of water, which can be used to fill a jug.
- The ground, onto which water can be poured away.

Neither jug has any markings, so you cannot read off a measurement such as "1 gallon" from the side of a jug. You can only know a jug is empty or completely full, or work out amounts by pouring between the jugs.

> **Question**
>
> How can you get exactly 2 gallons of water in the 4-gallon jug?

The difficulty is that 2 gallons cannot be measured directly. Neither jug is 2 gallons in size, so the 2 gallons has to be created indirectly, by filling, emptying and pouring between the jugs in a clever order.

3. Key Terms Explained in Simple Words

Term	Simple meaning
State	A snapshot of the situation at one moment: how much water is in each jug.
Initial state	The situation at the start. Here, both jugs are empty.
Goal state	The situation we want to reach. Here, exactly 2 gallons in the 4-gallon jug.
Operator / Action	One allowed step, such as "fill the 3-gallon jug" or "pour water from one jug into the other".
Production rule	A rule written as: IF a condition is true in the current state, THEN the action changes the state in a stated way.
State space	The set of all states that can possibly be reached by using the actions.
Search	Systematically exploring the state space to find a path from the initial state to the goal state.
Path cost	The price of a solution. Here every action costs 1, so cost equals the number of steps.
Optimal solution	A solution with the smallest possible number of steps.

4. Representing the Problem

4.1 State representation

A state is written as a pair of numbers (x, y):

- **x** = the quantity of water in the 4-gallon jug, so x can be 0, 1, 2, 3 or 4.

- **y** = the quantity of water in the 3-gallon jug, so y can be 0, 1, 2 or 3.

For example, (4, 0) means the 4-gallon jug is full and the 3-gallon jug is empty. (1, 3) means 1 gallon in the 4-gallon jug and the 3-gallon jug is full.

4.2 Initial state, goal state and size of the state space

Item	Definition
Initial state	(0, 0): both jugs are empty.
Goal state	(2, n): exactly 2 gallons in the 4-gallon jug. The amount in the 3-gallon jug, n, does not matter, so (2, 0), (2, 1), (2, 2) and (2, 3) are all accepted.
Total possible pairs	5 values of x × 4 values of y = 20 pairs.
Reachable from (0, 0)	Only 14 of the 20 pairs. The others, such as (1, 1), (1, 2), (2, 1), (2, 2), (3, 1) and (3, 2), can never occur using the allowed actions.

The grid below shows all 20 possible states. Green cells can be reached from the start, grey cells cannot, and the blue cells are goal states.

x \ y	y = 0	y = 1	y = 2	y = 3
x = 0	(0, 0)	(0, 1)	(0, 2)	(0, 3)
x = 1	(1, 0)	(1, 1)	(1, 2)	(1, 3)
x = 2	**(2, 0)**	(2, 1)	(2, 2)	**(2, 3)**
x = 3	(3, 0)	(3, 1)	(3, 2)	(3, 3)
x = 4	(4, 0)	(4, 1)	(4, 2)	(4, 3)

5. The Actions (Production Rules)

Every legal step is described by a production rule. Each rule says: if the current state (x, y) satisfies a condition, the new state will be the one shown. Rule numbers are used later to describe the solution.

No.	Condition (IF)	New state (THEN)	Meaning in plain words
1	x < 4	(4, y)	Fill the 4-gallon jug from the pump.
2	y < 3	(x, 3)	Fill the 3-gallon jug from the pump.

No.	Condition (IF)	New state (THEN)	Meaning in plain words
3	$x > 0$	$(0, y)$	Empty the 4-gallon jug on the ground.
4	$y > 0$	$(x, 0)$	Empty the 3-gallon jug on the ground.
5	$x + y \geq 4$ and $y > 0$	$(4, y - (4 - x))$	Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full.
6	$x + y \geq 3$ and $x > 0$	$(x - (3 - y), 3)$	Pour from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full.
7	$x + y \leq 4$ and $y > 0$	$(x + y, 0)$	Pour all the water from the 3-gallon jug into the 4-gallon jug (it all fits).
8	$x + y \leq 3$ and $x > 0$	$(0, x + y)$	Pour all the water from the 4-gallon jug into the 3-gallon jug (it all fits).

A note on the pouring rules. When pouring from one jug to another, you stop either when the source jug is empty or when the receiving jug is full, whichever happens first. Rules 5 and 7 cover both cases for pouring into the 4-gallon jug, and rules 6 and 8 cover both cases for pouring into the 3-gallon jug.

Example of applying a rule. Suppose the state is (3, 3). Rule 5 applies because $x + y = 6 \geq 4$ and $y > 0$. The 4-gallon jug needs $4 - 3 = 1$ more gallon to be full, so 1 gallon moves across and the new state is $(4, 3 - 1) = (4, 2)$.

Rules that pour away only a part of the water onto the ground are not included, because the quantity thrown away would be unmeasurable (the jugs have no markings) and such rules never help.

6. Searching for the Solution

6.1 Breadth-First Search (BFS)

Breadth-First Search explores the state space level by level. It first looks at every state that is 1 step from the start, then every state that is 2 steps away, then 3 steps, and so on. A state that has already been visited is never added again, which prevents going round in circles. Because shorter paths are checked first, the first time BFS reaches a goal state it has found a shortest solution.

Running BFS from the initial state (0, 0) gives the following levels:

Level (steps)	New states found	How they are produced
0	(0, 0)	The starting state.
1	(4, 0), (0, 3)	Fill the 4-gallon jug, or fill the 3-gallon jug.
2	(4, 3), (1, 3), (3, 0)	From (4, 0): fill the 3-gallon jug, or pour 4 into 3. From (0, 3): pour 3 into 4.
3	(1, 0), (3, 3)	Empty the 3-gallon jug from (1, 3). Fill the 3-gallon jug from (3, 0).
4	(0, 1), (4, 2)	Pour 4 into 3 from (1, 0). Pour 3 into 4 from (3, 3).
5	(4, 1), (0, 2)	Fill the 4-gallon jug from (0, 1). Empty the 4-gallon jug from (4, 2).
6	(2, 3), (2, 0)	Pour 4 into 3 from (4, 1). Pour 3 into 4 from (0, 2). GOAL FOUND.

In total BFS visits 14 states (1 + 2 + 3 + 2 + 2 + 2 + 2 = 14), which matches the 14 reachable states in the grid in Section 4. The goal state first appears at level 6, so the shortest solution needs exactly 6 steps. Two different shortest solutions exist, one ending at (2, 0) and one ending at (2, 3).

6.2 Other search methods

- **Depth-First Search (DFS):** follows one sequence of actions as far as possible before going back. It also finds a solution here, but the solution may be longer than necessary, and it needs a check for repeated states or it can loop forever (for example fill, empty, fill, empty...).
- **Heuristic search (A*):** is useful for large problems, but this state space has only 14 reachable states, so plain BFS is already fast and gives the best answer.

7. The Solutions, Step by Step

7.1 Solution 1 (ends at state (2, 0))

Idea: use the 3-gallon jug twice to put 6 gallons through the 4-gallon jug, getting rid of 4 of them, so 2 gallons remain.

Step	Action (rule applied)	State after	What is happening
0	Start	(0, 0)	Both jugs are empty.
1	Fill the 3-gallon jug (Rule 2)	(0, 3)	The 3-gallon jug is full.

Step	Action (rule applied)	State after	What is happening
2	Pour 3-gallon into 4-gallon (Rule 7)	(3, 0)	All 3 gallons fit, so the 4-gallon jug holds 3.
3	Fill the 3-gallon jug (Rule 2)	(3, 3)	Both jugs now hold 3 gallons each.
4	Pour 3-gallon into 4-gallon until full (Rule 5)	(4, 2)	The 4-gallon jug only had room for 1 more gallon, so 2 gallons stay in the 3-gallon jug.
5	Empty the 4-gallon jug (Rule 3)	(0, 2)	The 2 gallons we want are now in the 3-gallon jug.
6	Pour 3-gallon into 4-gallon (Rule 7)	(2, 0)	The 2 gallons move across. GOAL: 2 gallons in the 4-gallon jug.

7.2 Solution 2 (ends at state (2, 3))

Idea: fill the 4-gallon jug first and use the 3-gallon jug to take water away in pieces of 3 gallons.

Step	Action (rule applied)	State after	What is happening
0	Start	(0, 0)	Both jugs are empty.
1	Fill the 4-gallon jug (Rule 1)	(4, 0)	The 4-gallon jug is full.
2	Pour 4-gallon into 3-gallon until full (Rule 6)	(1, 3)	3 gallons move across, leaving 1 gallon behind.
3	Empty the 3-gallon jug (Rule 4)	(1, 0)	Only the 1 gallon in the 4-gallon jug remains.
4	Pour 4-gallon into 3-gallon (Rule 8)	(0, 1)	The 1 gallon moves to the 3-gallon jug.
5	Fill the 4-gallon jug (Rule 1)	(4, 1)	The 4-gallon jug is full again; the 3-gallon jug holds 1.
6	Pour 4-gallon into 3-gallon until full (Rule 6)	(2, 3)	The 3-gallon jug only had room for 2 gallons, so exactly 2 gallons are left in the 4-gallon jug. GOAL.

> **Which one should I write as the answer?**

> Both solutions are correct and both take 6 steps. Solution 1 is the classic textbook answer because it finishes with the 3-gallon jug empty, in the state (2, 0). Solution 2 is also valid because the goal only asks for 2 gallons in the 4-gallon jug.

7.3 The search path as a sequence of states

Solution 1: (0, 0) → (0, 3) → (3, 0) → (3, 3) → (4, 2) → (0, 2) → (2, 0)

Solution 2: (0, 0) → (4, 0) → (1, 3) → (1, 0) → (0, 1) → (4, 1) → (2, 3)

8. Checking the Solution

A good way to check a solution is to count the total water. In Solution 1:

- Water added by the pump: 3 gallons in step 1 and 3 gallons in step 3, so 6 gallons in total.
- Water thrown away on the ground: 4 gallons in step 5.
- Water left in the jugs at the end: 6 − 4 = 2 gallons, all in the 4-gallon jug. This matches the final state (2, 0).

In Solution 2, the pump adds 4 + 4 = 8 gallons, 3 gallons are thrown away in step 3, and 3 gallons remain in the 3-gallon jug at the end, so 8 − 3 = 5 gallons are in the jugs: 2 in the 4-gallon jug and 3 in the 3-gallon jug, which matches (2, 3). Every step also respects the limits: x never exceeds 4 and y never exceeds 3.

9. Why Does It Work? (Understanding the Idea)

The jugs hold 4 and 3 gallons. Every time we pour, fill or empty, the amount of water changes by 3 or 4 gallons (or moves between the jugs). Combining 3s and 4s can create new amounts:

- 4 − 3 = 1 gallon (this is how Solution 2 produces a single gallon).
- 3 + 3 − 4 = 2 gallons (this is exactly Solution 1: two fills of the 3-gallon jug, minus one full 4-gallon jug).
- 4 + 4 − 3 − 3 = 2 gallons (this is Solution 2: two fills of the 4-gallon jug, minus two 3-gallon portions).

Because 3 and 4 have no common factor other than 1, it is possible to measure any whole number of gallons from 1 to 4 using these two jugs. If the jug sizes did share a factor, some amounts would be impossible. For example, with a 4-gallon jug and a 6-gallon jug, every measurable amount is a multiple of 2, so you could never get exactly 3 gallons. This is the same idea as the parity rule for the 8-puzzle: some goals are simply unreachable, and it helps to know this before searching.

10. Real-World Relevance

- Planning and scheduling: many real tasks are sequences of allowed actions that must reach a target situation, just like pouring between jugs.
- Robotics and automation: a robot choosing steps to reach a desired position or arrangement solves a state-space search problem.
- Industry and chemistry: dosing and mixing exact quantities using only fixed-size containers uses the same logic.
- Games and puzzles: many puzzle solvers and game-playing programs use BFS, DFS and rule-based state transitions exactly as shown in this report.

11. Conclusion

The Water Jug Problem was formulated as a state-space search problem. A state is a pair (x, y) giving the water in the 4-gallon and 3-gallon jugs. The start state is (0, 0), the goal is any state of the form (2, n), and eight production rules describe the legal fill, empty and pour actions. Out of 20 possible pairs, 14 states are reachable from the start.

Breadth-First Search showed that the goal can be reached in a minimum of 6 steps. Two shortest solutions exist: (0, 0) → (0, 3) → (3, 0) → (3, 3) → (4, 2) → (0, 2) → (2, 0), and (0, 0) → (4, 0) → (1, 3) → (1, 0) → (0, 1) → (4, 1) → (2, 3). Either one gives exactly 2 gallons in the 4-gallon jug.

12. Summary of Results

Item	Value
State representation	(x, y): x in 4-gallon jug, y in 3-gallon jug
Initial state	(0, 0)
Goal state	(2, n), exactly 2 gallons in the 4-gallon jug
Total possible states	20
Reachable states	14
Production rules used	8
Search method	Breadth-First Search
Minimum number of steps	6

Item	Value
Solution 1	(0,0) → (0,3) → (3,0) → (3,3) → (4,2) → (0,2) → (2,0)
Solution 2	(0,0) → (4,0) → (1,3) → (1,0) → (0,1) → (4,1) → (2,3)